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STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION AND REQUEST FOR A THREE-YEAR ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

Approved By
Daniela Gressani and Bob Traa (IMF) and Bernard Funck and Jeffrey D. Lewis (IDA)

Prepared by the Staffs of the International Monetary Fund and the International Development Association

Risk of external debt distress:	**Moderate**
Augmented by significant risks stemming from domestic public debt and/or private external debt?	**Yes (domestic debt)**

This debt sustainability analysis (DSA) updates the joint IMF/IDA DSA from July 2, 2013. The results indicate that Yemen continues to face a moderate risk of debt distress. Although the level of public and publicly guaranteed external debt is modest, it is vulnerable to shocks to export receipts. Public domestic debt is large, leading to increased vulnerabilities, particularly to a growth shock.[1]

[1] This analysis is based on the joint Fund-Bank debt sustainability framework for conducting debt sustainability analysis in low-income countries. Under IDA's Country Policy and Institutional Assessment (CPIA), Yemen's three-year average score in 2010–12 is 3.04 and its policy performance is categorized as weak. This DSA update uses the indicative thresholds for countries for this category.

A. Underlying Assumptions

Apart from short-term projections, no major changes have been made to macroeconomic projections compared to the previous DSA (Box 1). Over the long term, real GDP growth is assumed to be about 4.5 percent, while exports and imports of goods and services are assumed to grow by about 5.5 percent and 3 percent, respectively. Inflation is assumed to stabilize at 6.5 percent. Projections for workers' remittances have been revised upwards in view of their strong performance in recent years, and LNG exports will be boosted by upward revisions to export prices in the contracts recently concluded or expected to be concluded.

B. External DSA

Under the baseline scenario, all public and publicly guaranteed (PPG) external debt indicators are below the policy-dependent thresholds over the projection period (2014–2034). The present value (PV) of debt-to-exports ratio is projected to peak at about 65 percent in 2026 before declining to 57 percent, compared to the threshold applied to Yemen (100 percent) (Table 1). In terms of the composition of public external debt, no contracting of commercial debt is assumed under the scenarios. However, about 40 percent of official debt is assumed to be semi-concessional (loans with a grant element of 25-35 percent) or reflecting obligations owed to the IMF and the Arab Monetary Fund.

Under the alternative scenarios and stress tests, all indicators remain below the thresholds, with one exception (Table 2 and Figure 1). The PV of debt-to-exports ratio is projected to rise above 100 percent in 2016 under the most extreme shock to export receipts and to remain so throughout the projection period. These findings highlight Yemen's high dependence on oil exports for its external sustainability.

Yemen's international investment position has not been published since 2008. Data on private sector flows and stocks, including nonhydrocarbon FDI and other private investment flows need to be improved. Existing shortcomings result in substantial unidentified flows that can be better classified if data improve.

C. Public DSA

Yemen's public debt-to-GDP ratio exceeds the policy-dependent benchmark in 2013[2] and remains above it through 2034 (Table 3). Even though the level of public and publicly-guaranteed external debt is comfortably low (about 8–16 percent), the PV of total public debt-to-GDP ratio amounts to 49 percent in 2013, exceeding the benchmark of 38 percent applied to Yemen. The public debt-to-GDP ratio increased

[2] Yemen's public debt at end-2013 was about 49 percent of GDP; 15 percentage points in external debt and 34 percentage points in domestic debt. All external debt is concessional, long term, and mostly denominated in US dollars. Domestic debt is all in local currency. About 25 percent of it is owed to the central bank in the form of loans that will be reduced by about 15 percent annually starting in 2017. About 30 percent is owed mostly in 5 years bonds to the corporate sector, insurance companies, and pension funds. Finally 45 percent is owed to the banking sector, most of which are in six months bills held by conventional banks, and only 5 percentage points are in 5–14 years sukuk held by Islamic banks.

Box 1. Assumptions for the Macroeconomic Framework

Growth: In the medium term (2014–19), real GDP growth is projected to average about 4.6 percent, which is slightly higher than the 10-year historical average before the political crisis (4.3 percent). This reflects projected recovery from the recent economic slowdown because of the supply shock, increased international assistance, including the proposed ECF arrangement, and utilization of idle production capacity. For the long term growth is projected to average 4.5 percent (2020–34). Given the gradual decline in oil production, non-hydrocarbon growth is essential. Following the political transition and under the ECF-supported program, the government will undertake structural reforms to improve the business environment and attract sizeable donor-funded public investment. Reprioritization of expenditures should support infrastructure investment, while reduction in crowding out would lead to growth in credit to the private sector, boosting non-hydrocarbon growth. The assumptions on nonhydrocarbon growth rates average 5.2 percent in 2014–19 and 4.7 percent in the long term. These are broadly in line with the historical experience.

Hydrocarbon Sector: As in last year's DSA, oil production is assumed to continue towards the end of projection period, in line with the authorities' projections and consistent with the assessment of main credible independent sources (e.g. British Petroleum Oil Statistics, *Oil & Gas Journal*, and Economic Intelligence Unit), which put Yemen's proven oil reserves at about 3 billion barrels. Assuming a partial recovery in oil production over the medium term, and subsequent declines of 3 percent per year over the long-term, Yemen would use only about half of its proven reserves by 2034. Yemen has a large potential in the LNG sector. The DSA assumes no increase in LNG production over the medium-term and an average annual increase at 7 percent from 2025 onwards.

Inflation: Inflation is projected to edge up in 2014–15 owing to the impact of higher fuel prices following reforms, and to gradually decline to single digits in the medium term with the recovery of economic activities. It would then moderate to an average of 6.5 percent in the long term, assuming that the central bank refrains from financing the government budget and ensures the stability of the real exchange rate.

Public finance: Despite the pressure on the fiscal position following the shock to oil production in early 2014, strong fiscal adjustment will be undertaken under the ECF-supported program and will continue at a more gradual pace over the projection period. The ratio of total revenues (excluding grants) to GDP would thus hover around 20 percent. On the expenditure side, the government's efforts to control the public wage bill and fuel subsidies, including by removing ghost workers and double dippers as well as increasing domestic petroleum prices, are expected to lead to a reduction in overall expenditures from 29.6 percent of GDP in 2014 to 22.1 percent in 2034. As a result, the overall fiscal deficit would gradually decline to around 3 percent of GDP by 2034.

External sector: Exports are expected to decrease in the medium term, reflecting a declining oil production. However, a gradual increase in non-oil exports and, more importantly, LNG exports, will offset the decrease in oil exports in the long run. The value of LNG exports will be boosted by higher LNG prices on account of recently renegotiated LNG export contracts. Nonhydrocarbon export growth would be around 6.4 percent in 2015–19, and would stabilize at around 5.5 percent in the long run. Import growth is projected at a conservative average of 2.9 percent over the projection period, reflecting fiscal adjustment. The current account deficit would gradually increase to about 2 percent of GDP in the medium term and would slowly widen over the projection period to about 3–4 percent of GDP. Official creditors are expected to continue providing financing on favorable terms. Improved security conditions and business environment would attract more foreign direct investment.

by 12 percentage point of GDP from 2008 to 2013. Most of the debt remained denominated in local currency. The maturity of domestic debt is shorter than external debt. The creditor base is domestic and very narrow. Key holders of domestic debt are commercial and Islamic banks, the Pension Fund, and a telecommunication company. These facts suggest substantial rollover and interest rate risks when facing a shock. The expansion of domestic public debt also resulted from financing of the budget by the central bank in the aftermath of the 2011 crisis. Data on contingent liabilities are not available, but there is some anecdotal evidence that indicates the existence of such liabilities.

The alternative scenarios and stress tests point to the importance for Yemen to embark on fiscal consolidation (Table 4 and Figure 2). They suggest that Yemen's public debt position is particularly vulnerable to GDP growth shocks.

D. Conclusion

Based on an assessment of PPG external debt, Yemen continues to face a moderate risk of debt distress. The assessment of the risk is reinforced by significant vulnerabilities related to a high level of public domestic debt. While the debt profile remains sustainable in a scenario of recovery and stability under continued prudent policies, it is subject to substantial downside risks, including a deterioration of the security environment and a decline in oil production and exports.

The authorities broadly agreed with staff's assessment. However, they highlighted the downside risks, especially in view of the potential impact of the projected decline in oil production, and the risk of deterioration in the fiscal and external accounts in case of slippages in policy implementation that might result from deterioration in the security situations.

Table 1. Republic of Yemen: External Debt Sustainability Framework, Baseline Scenario, 2011–34

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**19.9**	**18.7**	**16.4**			**16.4**	**16.5**	**17.6**	**18.5**	**19.0**	**19.7**		**20.2**	**16.3**	
of which: public and publicly guaranteed (PPG)	18.6	17.4	15.2			13.7	13.2	13.7	14.0	14.2	14.4		13.2	9.8	
Change in external debt	-0.3	-1.2	-2.3			0.0	0.1	1.2	0.8	0.6	0.7		-0.1	0.0	
Identified net debt-creating flows	4.9	-0.8	2.5			5.0	2.8	1.7	1.4	1.4	1.7		3.5	3.7	
Non-interest current account deficit	**2.8**	**1.5**	**2.9**	**2.3**	**4.2**	**1.1**	**0.8**	**1.1**	**1.0**	**1.3**	**2.1**		**4.4**	**4.2**	3.9
Deficit in balance of goods and services	2.4	11.6	7.8			5.0	1.7	2.0	2.4	2.9	4.1		5.7	3.3	
Exports	30.3	24.9	22.4			23.3	24.0	23.4	22.7	21.9	20.4		16.1	14.7	
Imports	32.7	36.5	30.2			28.3	25.7	25.4	25.1	24.8	24.4		21.8	18.0	
Net current transfers (negative = inflow)	-6.5	-14.4	-9.2	-8.4	2.5	-8.7	-7.6	-7.2	-6.9	-6.6	-6.4		-5.1	-2.9	-4.4
of which: official	-2.4	-6.2	-0.9			-1.4	-1.0	-0.9	-0.9	-0.9	-0.9		-0.7	-0.4	
Other current account flows (negative = net inflow)	6.9	4.2	4.3			4.8	6.8	6.4	5.6	5.0	4.5		3.7	3.7	
Net FDI (negative = inflow)	**3.1**	**-1.0**	**1.8**	**-0.6**	**3.1**	**3.9**	**2.3**	**1.1**	**0.8**	**0.7**	**0.1**		**-0.6**	**-0.2**	-0.4
Endogenous debt dynamics 2/	**-0.9**	**-1.3**	**-2.1**			**-0.1**	**-0.4**	**-0.4**	**-0.5**	**-0.6**	**-0.5**		**-0.3**	**-0.2**	
Contribution from nominal interest rate	0.2	0.2	0.2			0.2	0.2	0.3	0.4	0.4	0.5		0.6	0.5	
Contribution from real GDP growth	2.4	-0.4	-0.8			-0.3	-0.7	-0.7	-0.9	-1.0	-1.0		-0.8	-0.7	
Contribution from price and exchange rate changes	-3.6	-1.1	-1.5			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-5.2**	**-0.5**	**-4.8**			**-5.0**	**-2.7**	**-0.6**	**-0.6**	**-0.8**	**-1.0**		**-3.6**	**-3.7**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	17.0			17.1	17.3	18.1	18.5	18.7	18.9		18.7	14.7	
In percent of exports	...	...	76.1			73.4	72.0	77.2	81.5	85.5	92.8		115.9	99.8	
PV of PPG external debt	**...**	**...**	**15.8**			**14.3**	**14.0**	**14.2**	**14.1**	**13.8**	**13.7**		**11.7**	**8.2**	
In percent of exports	**...**	**...**	**70.5**			**61.7**	**58.5**	**60.5**	**62.0**	**63.2**	**67.0**		**72.4**	**55.8**	
In percent of government revenues	**...**	**...**	**68.6**			**63.6**	**64.7**	**64.0**	**63.2**	**64.2**	**63.5**		**55.8**	**43.4**	
Debt service-to-exports ratio (in percent)	**2.8**	**2.9**	**3.0**			**4.0**	**3.9**	**4.4**	**5.8**	**5.4**	**6.0**		**8.9**	**5.5**	
PPG debt service-to-exports ratio (in percent)	**2.8**	**2.9**	**2.9**			**4.0**	**3.8**	**4.1**	**5.2**	**4.6**	**4.8**		**6.9**	**3.6**	
PPG debt service-to-revenue ratio (in percent)	**3.5**	**3.0**	**2.9**			**4.1**	**4.2**	**4.3**	**5.3**	**4.7**	**4.6**		**5.3**	**2.8**	
Total gross financing need (Billions of U.S. dollars)	2.2	0.4	2.1			2.7	2.1	1.7	1.8	1.9	2.2		4.4	7.3	
Non-interest current account deficit that stabilizes debt ratio	3.1	2.7	5.1			1.1	0.8	0.0	0.2	0.7	1.5		4.5	4.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-12.7	2.4	4.8	2.6	5.6	1.9	4.6	4.7	5.2	5.9	5.6	4.6	4.3	4.6	4.5
GDP deflator in US dollar terms (change in percent)	21.3	5.6	8.9	10.8	8.8	10.4	7.2	0.1	0.7	0.5	0.2	3.2	1.3	1.7	1.5
Effective interest rate (percent) 5/	1.2	1.1	1.1	1.2	0.3	1.6	1.7	1.9	2.1	2.4	2.6	2.0	2.9	3.0	2.9
Growth of exports of G&S (US dollar terms, in percent)	7.1	-11.4	2.7	9.7	20.6	16.9	15.8	2.1	2.7	2.6	-1.4	6.5	2.3	5.5	3.8
Growth of imports of G&S (US dollar terms, in percent)	0.7	20.7	-5.7	11.4	14.5	5.5	1.7	3.7	4.8	5.1	4.3	4.2	4.1	3.5	3.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	39.1	37.9	39.6	39.7	39.9	39.9	39.4	40.2	40.2	40.2
Government revenues (excluding grants, in percent of GDP)	24.1	23.8	23.0			22.5	21.7	22.1	22.3	21.6	21.5		21.0	18.9	20.4
Aid flows (in Billions of US dollars) 7/	0.5	2.4	0.6			0.8	0.9	0.9	1.0	1.2	1.2		1.2	1.3	
of which: Grants	0.4	2.2	0.4			0.6	0.5	0.4	0.4	0.5	0.5		0.4	0.5	
of which: Concessional loans	0.1	0.3	0.2			0.2	0.4	0.5	0.6	0.7	0.7		0.8	0.8	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.7	1.7	1.5	1.6	1.5	1.4		1.0	0.6	0.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			76.1	61.0	57.3	56.2	60.2	59.3		58.1	59.0	58.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	32.7	35.4	40.4			45.5	51.0	53.4	56.6	60.2	63.7		84.0	154.2	
Nominal dollar GDP growth	5.9	8.2	14.2			12.5	12.1	4.8	5.9	6.4	5.8	7.9	5.7	6.4	6.1
PV of PPG external debt (in Billions of US dollars)			6.4			6.5	6.9	7.3	7.7	8.1	8.5		9.6	12.3	
(PVt-PVt-1)/GDPt-1 (in percent)						0.4	0.9	0.8	0.7	0.7	0.6	0.7	0.2	0.3	0.3
Gross workers' remittances (Billions of US dollars)	1.4	3.0	3.4			3.4	3.4	3.5	3.5	3.6	3.6		3.9	4.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	14.5			13.3	13.1	13.3	13.3	13.1	12.9		11.2	8.0	
PV of PPG external debt (in percent of exports + remittances)	...	...	51.1			46.6	45.6	47.4	48.7	49.8	52.4		56.3	47.0	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.1			3.0	2.9	3.2	4.1	3.6	3.8		5.4	3.0	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Republic of Yemen: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	14	14	14	14	14	14	**12**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	14	13	12	11	10	9	**2**	-7
A2. New public sector loans on less favorable terms in 2014-2034 2	14	14	15	15	15	15	**15**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	14	15	16	16	16	16	**13**	9
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	14	17	22	21	21	21	**18**	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	14	14	14	14	14	14	**12**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	14	15	17	16	16	16	**14**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	14	16	19	18	18	18	**16**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	14	19	19	19	19	19	**16**	11
PV of debt-to-exports ratio								
Baseline	61	58	61	62	63	67	**72**	56
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	61	54	51	49	47	46	**12**	-47
A2. New public sector loans on less favorable terms in 2014-2034 2	61	58	62	66	69	75	**92**	86
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	61	57	59	60	62	65	**71**	54
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	61	90	136	139	142	150	**167**	106
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	61	57	59	60	62	65	**71**	54
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	61	61	71	73	74	79	**86**	60
B5. Combination of B1-B4 using one-half standard deviation shocks	61	73	90	92	94	100	**110**	75
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	61	57	59	60	62	65	**71**	54
PV of debt-to-revenue ratio								
Baseline	64	65	64	63	64	63	**56**	43
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	64	60	54	50	48	44	**10**	-37
A2. New public sector loans on less favorable terms in 2014-2034 2	64	64	66	67	70	71	**71**	67
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	64	67	72	71	73	72	**63**	49
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	64	78	98	96	97	96	**87**	55
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	64	66	64	63	64	64	**56**	43
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	64	68	75	74	75	74	**66**	47
B5. Combination of B1-B4 using one-half standard deviation shocks	64	74	84	83	84	83	**74**	51
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	64	87	86	85	87	86	**75**	58

Table 2. Republic of Yemen: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	4	4	4	5	5	5	**7**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	4	4	4	4	4	**3**	-1
A2. New public sector loans on less favorable terms in 2014-2034 2	4	4	4	5	5	5	**8**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	4	4	5	5	5	**7**	4
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	5	7	9	8	8	**12**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	4	4	5	5	5	**7**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	4	4	6	5	5	**7**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	5	7	6	6	**9**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	4	4	4	5	5	5	**7**	4
Debt service-to-revenue ratio								
Baseline	4	4	4	5	5	5	**5**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	4	4	4	4	3	**2**	-1
A2. New public sector loans on less favorable terms in 2014-2034 2	4	4	4	6	5	5	**6**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	4	5	6	5	5	**6**	3
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	4	5	6	5	5	**6**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	4	4	5	5	5	**6**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	4	4	6	5	5	**6**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	5	6	5	5	**6**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	4	6	6	7	6	6	**7**	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	40	40	40	40	40	40	**40**	40

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

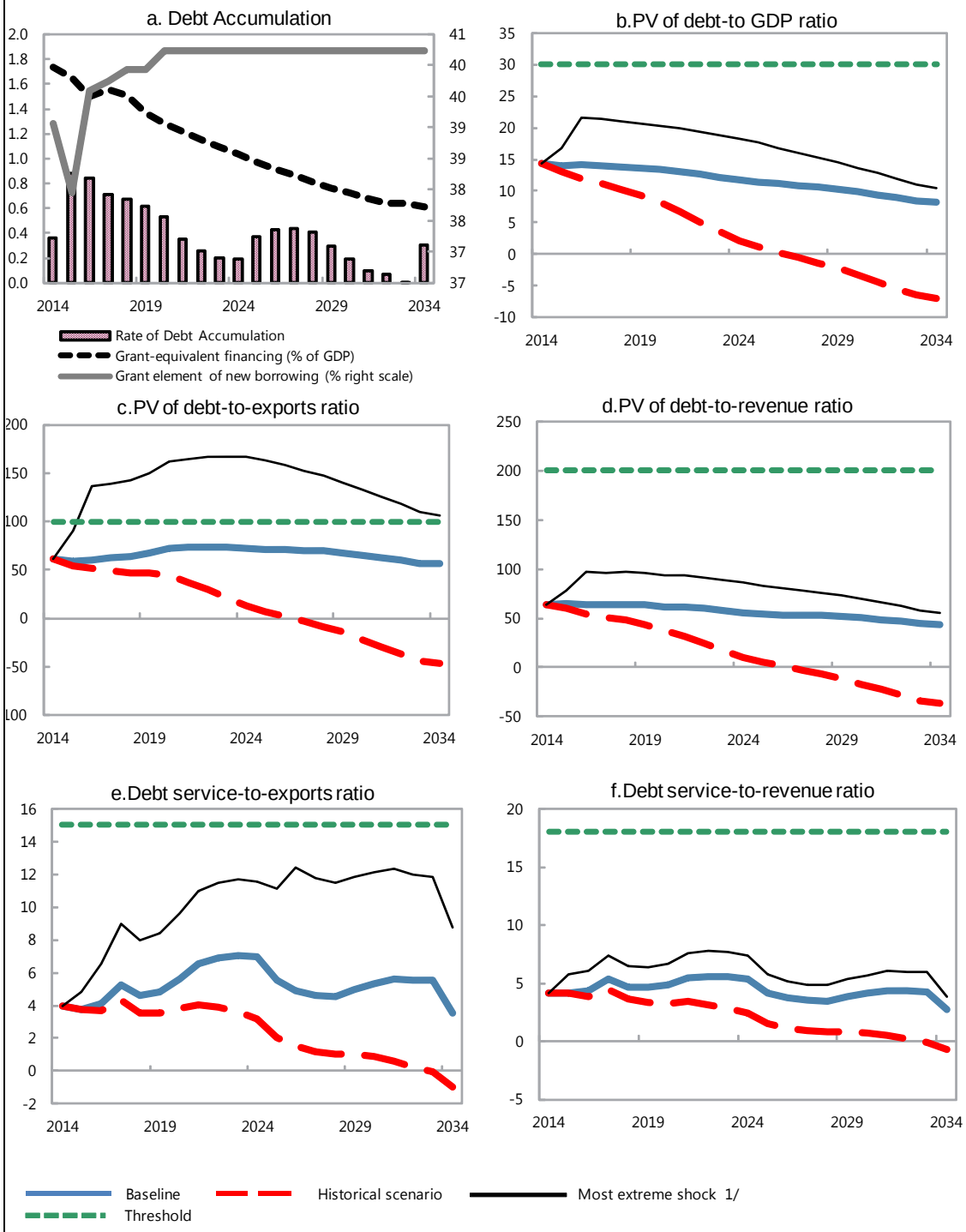
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Republic of Yemen: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock.

Table 3. Republic of Yemen: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections									
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	45.7	47.3	48.2			48.2	47.4	48.0	47.8	47.4	46.9		42.2	30.4	
of which: foreign-currency denominated	18.6	17.4	15.2			13.7	13.2	13.7	14.0	14.2	14.4		13.2	9.8	
Change in public sector debt	3.3	1.6	0.8			0.0	-0.8	0.6	-0.2	-0.5	-0.5		-1.3	-1.1	
Identified debt-creating flows	3.2	2.8	0.9			0.0	-0.8	0.6	-0.2	-0.5	-0.3		-1.1	-0.7	
Primary deficit	0.2	0.9	1.5	0.7	3.4	0.1	0.0	-0.2	-0.6	-0.4	-0.5	-0.3	-1.1	-0.6	-0.9
Revenue and grants	25.3	29.9	23.9			23.9	22.7	22.9	23.0	22.4	22.3		21.5	19.2	
of which: grants	1.2	6.1	0.9			1.4	1.0	0.8	0.8	0.8	0.7		0.5	0.3	
Primary (noninterest) expenditure	25.5	30.8	25.4			24.0	22.7	22.6	22.4	22.1	21.8		20.4	18.6	
Automatic debt dynamics	3.1	2.0	-0.5			0.0	-0.8	0.8	0.4	-0.1	0.2		0.1	-0.1	
Contribution from interest rate/growth differential	6.1	2.6	0.6			1.2	-0.6	0.7	0.2	-0.3	0.0		0.0	-0.2	
of which: contribution from average real interest rate	-0.1	3.7	2.8			2.1	1.5	2.8	2.6	2.4	2.5		1.8	1.2	
of which: contribution from real GDP growth	6.2	-1.1	-2.2			-0.9	-2.1	-2.1	-2.4	-2.6	-2.5		-1.8	-1.4	
Contribution from real exchange rate depreciation	-3.0	-0.6	-1.2			-1.2	-0.2	0.2	0.2	0.2	0.2		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.1	-1.3	-0.1			0.0	0.0	0.0	0.0	0.0	-0.2		-0.2	-0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	48.7			48.8	48.2	48.4	47.9	47.1	46.1		40.7	28.8	
of which: foreign-currency denominated	...	...	15.8			14.3	14.0	14.2	14.1	13.8	13.7		11.7	8.2	
of which: external	...	...	15.8			14.3	14.0	14.2	14.1	13.8	13.7		11.7	8.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.1	6.8	7.4			6.7	6.3	6.0	5.8	5.5	5.2		4.0	3.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	203.9			204.1	212.3	211.5	208.1	209.9	207.1		189.3	150.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	212.3			216.7	222.2	218.9	215.2	218.1	214.3		194.1	152.8	
of which: external 3/	...	...	68.6			63.6	64.7	64.0	63.2	64.2	63.5		55.8	43.4	
Debt service-to-revenue and grants ratio (in percent) 4/	19.6	19.9	24.8			27.6	27.5	27.4	27.8	26.2	25.7		24.0	19.8	
Debt service-to-revenue ratio (in percent) 4/	20.5	25.0	25.8			29.3	28.8	28.4	28.8	27.2	26.6		24.6	20.2	
Primary deficit that stabilizes the debt-to-GDP ratio	-3.2	-0.7	0.6			0.0	0.8	-0.8	-0.4	0.1	0.0		0.1	0.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-12.7	2.4	4.8	2.6	5.6	1.9	4.6	4.7	5.2	5.9	5.6	4.6	4.3	4.6	4.5
Average nominal interest rate on forex debt (in percent)	2.4	2.3	2.3	2.2	0.3	3.2	3.4	3.3	3.2	3.1	3.0	3.2	3.1	3.1	3.1
Average real interest rate on domestic debt (in percent)	-0.5	13.5	9.4	3.8	9.7	5.6	4.0	8.0	7.5	7.0	7.5	6.6	5.7	5.4	5.8
Real exchange rate depreciation (in percent, + indicates depreciation)	-13.7	-3.4	-7.1	-7.3	8.6	-8.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	18.1	5.9	9.2	12.6	9.2	10.4	11.0	6.6	6.8	6.4	5.5	7.8	6.4	6.8	6.6
Growth of real primary spending (deflated by GDP deflator, in percent)	-19.9	23.6	-13.6	-1.0	11.2	-3.4	-1.1	4.3	4.2	4.1	4.2	2.1	2.7	3.8	3.4
Grant element of new external borrowing (in percent)	...	...	...	...	...	39.1	37.9	39.6	39.7	39.9	39.9	39.4	40.2	40.2	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

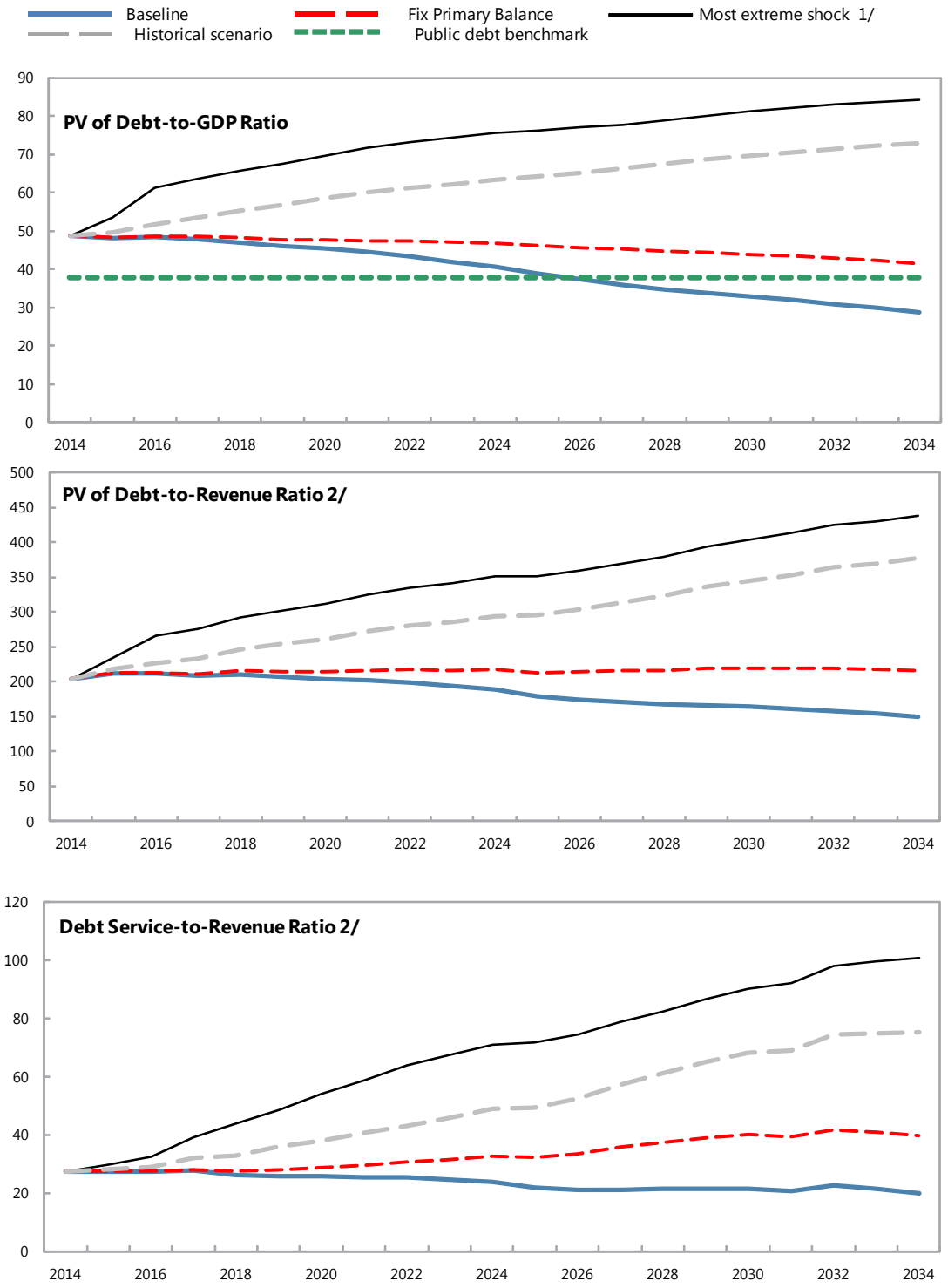
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Republic of Yemen: Sensitivity Analysis for Key Indicators of Public Debt 2014–34

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	49	48	48	48	47	46	41	29
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	49	50	52	54	55	57	63	73
A2. Primary balance is unchanged from 2014	49	48	49	49	48	48	47	42
A3. Permanently lower GDP growth 1/	49	49	50	51	52	52	59	83
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	49	54	61	64	66	68	76	84
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	49	52	56	55	54	53	47	34
B3. Combination of B1-B2 using one half standard deviation shocks	49	53	58	59	60	61	63	64
B4. One-time 30 percent real depreciation in 2015	49	55	55	54	53	52	46	36
B5. 10 percent of GDP increase in other debt-creating flows in 2015	49	57	58	57	56	54	48	35
PV of Debt-to-Revenue Ratio 2/								
Baseline	204	212	212	208	210	207	189	150
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	204	219	226	233	246	255	293	377
A2. Primary balance is unchanged from 2014	204	213	213	212	215	215	217	216
A3. Permanently lower GDP growth 1/	204	216	219	221	230	234	272	433
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	204	235	266	276	292	302	351	439
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	204	229	245	241	242	239	220	179
B3. Combination of B1-B2 using one half standard deviation shocks	204	232	252	257	267	272	294	332
B4. One-time 30 percent real depreciation in 2015	204	242	239	234	235	232	216	188
B5. 10 percent of GDP increase in other debt-creating flows in 2015	204	253	251	246	248	244	225	183
Debt Service-to-Revenue Ratio 2/								
Baseline	28	28	27	28	26	26	24	20
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	28	29	32	33	36	49	75
A2. Primary balance is unchanged from 2014	28	28	27	28	27	28	33	40
A3. Permanently lower GDP growth 1/	28	28	28	30	30	31	45	99
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	28	30	33	39	44	49	71	101
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	28	28	29	42	43	38	33	27
B3. Combination of B1-B2 using one half standard deviation shocks	28	29	31	39	40	43	55	71
B4. One-time 30 percent real depreciation in 2015	28	29	30	32	31	31	34	33
B5. 10 percent of GDP increase in other debt-creating flows in 2015	28	28	32	57	35	46	33	28

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure 2. Republic of Yemen: Indicators of Public Debt Under Alternative Scenarios, 2014-2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.